Model Question Paper

TED(15) 3064 (Revision 2015)

THIRD SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOFY

YARN FORMING – I

Time : 3 hrs Maximum marks: 100)

PART - A

(Maximum marks = 10 marks)

Marks

I Answer the following questions in one or two sentences - each carry 2 marks

1. Define ginning
2. State the function of evener roller in Hopper bale breaker
3. List the different contaminants in cotton fiber
4. Mention the heal and tow arrangement of flats
5. Define mechanical draft

PART – B

(Maximum marks = 30)

II. Answer any 5 questions – all questions carry equal marks

a) Distinguish different methods of mixing
b) Write various parts of the Mixing bale opener
c) List the various lap defects and discuss the causes
d) List the various types of waste in carding and mention a normal percentage of extraction
e) Illustrate Taker-in region of a card
f) Discuss the Principle of Tandem Carding
g) Find the actual production of the card per 8 hours in Kilograms from the following data Speed: 20 MPM, Hank sliver: 0.145, Efficiency: 90 %

PART – C

(Maximum marks = 60)

Answer one full question from each module

Module I

III a) With the help of a neat diagram describe the working of Auto Mixer (10)
b) State the function and working of swing door in Bale opener (5)

OR

IV a) Illustrate the working of Blendomat with the help of a sketch (10)
b) Suggest typical mixing plans for medium and fine cotton counts (5)

Module II

V. a) With the help of a neat diagram explain the working of Axi-flo cleaner (10)

b) State the functions of Dust filter bag and magnetic trap (5)

OR

VI. a) Find the actual Blow room production per shift of 8 hours in kilograms (10)

Bottom calendar roll speed	: 10 rpm	Bottom calendar roll diameter	: 7 inches
Lap length	: 44 meters	Lap weight	: 12 kilograms
Efficiency	: 82 %	Number of scutchers	: 2

b) Find the efficiency of the blow room (5)

Actual production/8 hour	: 980 kgs
Bottom calendar roller speed	: 8 rpm
Bottom calendar roller diameter	: 7 inches
Hank lap	: 0.00145

Module III

VII. a) Illustrate the working of H & B Flexible Bend with diagram (10)

b) Mention the functions of flexible bend and front plate in a card (5)

OR

VIII. a) Explain the maintenance operations given to a card (10)

b) Define trailing and leading hooks in card sliver (5)

Module IV

IX. a) Describe the working of Short term auto levellers employed in carding with diagram (10)

b) List the features of LC-363 HP card (5)

OR

X. Draw the draft gearing of the card and determine the Draft constant (15)

Lap roller wheel	: 48 T	Feed roll wheel	: 17 T
Feed roll plate wheel	: 120 T	Side shaft bevel	: 22 T
Doffer bevel	: 22 T	Doffer wheel	: 180 T
Calendar roll end wheel	: 16 T	Calendar roll off end wheel	: 25 T
Canon shaft wheel	: 16 T	Canon shaft bevel	: 170 T
Upright shaft middle bevel	: 17 T	Upright shaft top bevel	: 17 T
Coiler calendar roll bevel	: 17 T	Lap roller diameter	: 6 inches
Coiler calendar roll diameter	: 2 inches		

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